

Design Automation of a Fully Differential Switched Capacitor Amplifier Using a Telescopic Cascode OTA

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ABSTRACT

In this paper, we propose a design procedure and an automated design flow for a fully differential switched capacitor amplifier. The proposed procedure uses the gm/ID design methodology, which relies on searching lookup tables pre-generated from accurate simulation models. A set of specifications including power consumption, noise, bandwidth, gain, static error, and output swing is used to constrain the design process. The operational transconductance amplifier (OTA) is implemented using a telescopic cascode architecture. The automation program is written in MATLAB and is capable of generating sizing parameters, writing circuit netlists, running simulations, and displaying and analyzing simulation results. A design example is used to verify the proposed program, where Cadence Spectre is used to verify the achievement of the required circuit specifications.

Keywords: Switched-capacitor amplifier; telescopic cascode OTA; analog design automation; g_m/I_D design methodology; CMOS

I. INTRODUCTION

The microelectronics market is dominated by CMOS digital microprocessors and memories; however, analog/mixed-signal circuits will always be needed to link between the digital electronic circuits and the physical analog world. Advances in electronic design automation (EDA) enabled the automation of almost all the digital integrated circuits design flow. On the other hand, analog automation is still lagging behind. The design of CMOS analog integrated circuits not only requires lots of experience and intuition, but also long time-consuming iterations on expensive simulation tools. In order to achieve a set of required circuit specifications, the designer should be able to describe the devices used with proper and reliable models. However, on one hand, the simple MOSFET square law model fails to describe devices in moderate/weak inversion, in addition to short channel devices. On the other hand, accurate models for such devices are too complicated to be used in hand analysis. An attractive solution for this dilemma is using the gm/ID design methodology [1] [2]. The gm/ID method depends on using pre-generated lookup tables of MOSFET parameters obtained from a set of DC and noise simulations. This approach enables accurate design values inherited from the accuracy of the lookup tables. Moreover, the lookup tables enable design optimization and exploring the different trade-offs between circuit parameters.

Although the gm/ID design methodology offers an accurate and a reliable solution, the design procedure of an analog block still needs significant time and effort, which could fortunately be saved by automating the design flow using a computer program. This paper proposes the design automation of a fully-differential switched capacitor (SC) amplifier, which finds many applications in analog to digital converters (ADCs), analog filters, signal conditioning circuits, and sensor readout circuits [2] [3]. The operational transconductance amplifier (OTA) is implemented using a telescopic cascode architecture, which –although suffering from limited swing– offers superior energy efficiency compared to folded cascode and two-stage OTAs [4] [5]. The design procedure is constrained by several circuit specifications. The proposed automation flow is verified using Cadence Spectre in 0.18 μ m technology [6], where the results are in close agreement with the required specifications.

The rest of the paper is organized as follows: Section II describes the specifications of the switched capacitor amplifier, Section III provides circuit analysis and describes the equations used in the automated design flow, Section IV explains the design procedure, and Section V presents the automation and verification results.

II. SWITCHED CAPACITOR AMPLIFIER SPECIFICATIONS

The schematic of the fully-differential SC amplifier is shown in Fig. 1(a), and the schematic of the telescopic cascode OTA is shown in Fig. 1(b).

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Without loss of generality, a set specifications is assumed for the proposed design example as shown in Table 1. The proposed procedure can be similarly applied to other specifications. Although the 0.18 μ m node chosen for this design example has been succeeded by several generations, it is still popular for the implementation of analog/mixed signal systems that do not require high speed operation or high transistor count [5] [7]. The OTA is implemented using a PMOS input pair. A reference current (I_{REF}) is assumed to be externally provided.

Table 1: Specifications of the switched capacitor amplifier

Technology	0.18 μ m
Supply Voltage (V_{DD})	1.8 V
OTA Bias Current (I_{BIAS})	20 μ A
Load Capacitance (C_L)	2 pF
DC Closed-Loop Gain ($A_{V_{CLo}}$)	2
Closed-Loop Bandwidth (BW_{CL})	3 MHz
RMS Input-Referred Noise Voltage ($V_{n,i,rms}$)	85 μ Vrms
Static-Gain Error (ϵ_s)	0.15%
Differential Output Swing (V_{swing})	1.2 V_{pk-pk}

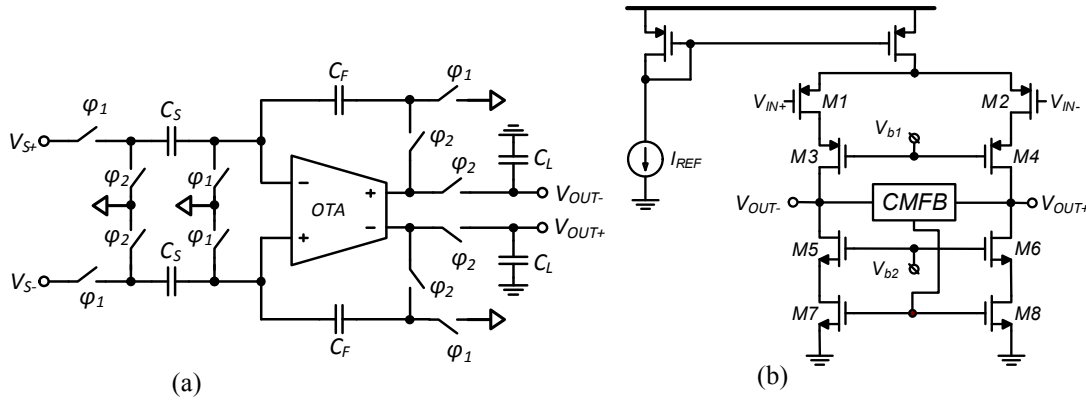


Fig. 1: (a) Schematic of the switched capacitor circuit and (b) the telescopic cascode OTA.

III. CIRCUIT ANALYSIS

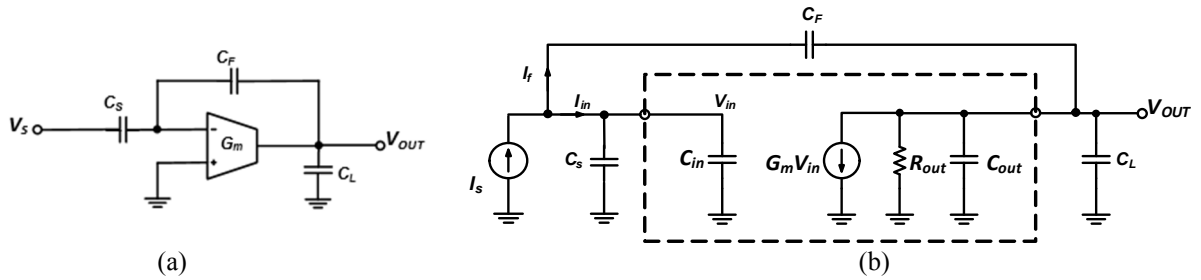


Fig. 2: (a) SC amplifier half-circuit in amplification phase and (b) Norton equivalent circuit.

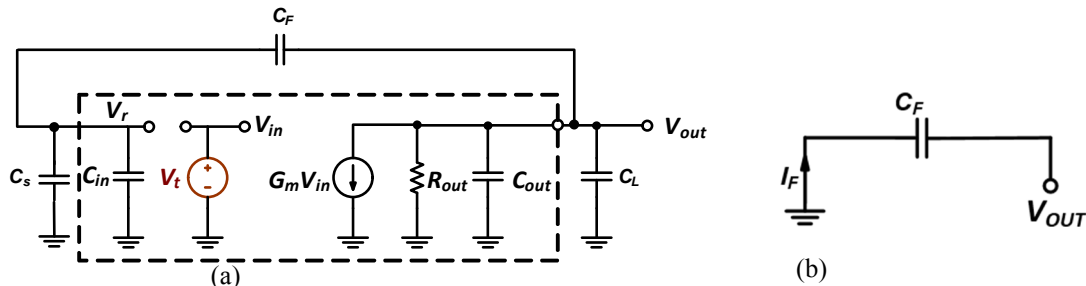


Fig. 3: (a) Breaking the loop to calculate the loop gain and (b) feedback network of the shunt-shunt amplifier.

The operation of the circuit in Fig. 1(a) starts in the sampling phase ($\phi_1 = 1, \phi_2 = 0$), where a sample of the input signal (V_s) is stored on the pair of sampling capacitors (C_s). Next, in the amplification (charge redistribution) phase ($\phi_1 = 0, \phi_2 = 1$), the sampled signal acts as an input signal for a fully differential amplifier, where the charge is transferred from C_s to the pair of feedback capacitors (C_F). Due to the symmetry of the fully differential circuit, the analysis can be simplified by using the half-circuit principle. Fig. 2(a) shows the equivalent half circuit in the amplification phase. Since the output voltage (V_{out}) is sampled and fed back as a current signal (I_f), the circuit can be analyzed as a shunt-shunt feedback amplifier (voltage-sampling current-mixing feedback). For a shunt-shunt feedback circuit, the input signal is better represented by a current source (I_s) to match the type of the feedback signal. Fig. 2(b) shows Norton equivalent circuit where I_s is given by

$$I_s = sC_s V_s \quad (1)$$

and the error signal (I_{in}) is equal to $I_s - I_f$. The OTA model could be well approximated as a single transconductance stage as shown in Fig. 2(b). The transconductance (G_m), output resistance (R_{out}), and input capacitance (C_{in}) of the OTA are approximately given by

$$G_m \approx g_{m1,2} \quad (2)$$

$$R_{out} \approx g_{m3,4} r_{o3,4} r_{o1,2} // g_{m5,6} r_{o5,6} r_{o7,8} \quad (3)$$

$$C_{in} \approx C_{gs1,2} + C_{gd1,3} * \left(1 + g_{m1,2} \left(r_{o1,2} || \frac{1}{g_{m3,4}} \right) \right) \approx C_{gs1,2} + 2C_{gd34} \quad (4)$$

where we assume that $\frac{1}{g_{m3,4}} \ll r_{o1,2}$ and $\frac{g_{m1,2}}{g_{m3,4}} \approx 1$. Moreover, the parasitic capacitance at the OTA output (C_{out}) can be neglected compared to the large load capacitance.

The loop gain (LG) of the feedback amplifier can be calculated by breaking the feedback loop and applying a test source (V_t) as shown in Fig. 3(a). The LG is given by

$$LG = \beta A = -\frac{V_r}{V_t} = \frac{\beta G_m R_{out}}{1 + s R_{out} C'_{out}} \quad (5)$$

where C'_{out} is the total output capacitance that is given by

$$C'_{out} = C_{out} + C_L + \frac{C_F(C_s + C_{in})}{C'_{in}} \approx C_L + C_F (1 - \beta) \quad (6)$$

where β is the series-shunt feedback factor which is given by

$$\beta = \frac{C_F}{C'_{in}} \quad (7)$$

and C'_{in} is total input capacitance that is given by

$$C'_{in} = C_s + C_F + C_{in} \quad (8)$$

Noting that the loop gain of the shunt-shunt and series-shunt configurations is the same because it is independent of the input signal type and location, the transimpedance open-loop gain is given by

$$A_{ROL} = \left(\frac{V_{out}}{I_s} \right)_{OL} = \frac{LG}{\beta_G} = \frac{\beta}{\beta_G} \cdot \frac{G_m R_{out}}{1 + s R_{out} C'_{out}} \quad (9)$$

where β_G is the transimpedance feedback factor which can be calculated from Fig. 3(b) using the shunt-shunt approximation as follows

$$\beta_G = \frac{V_{out}}{I_f} = -sC_F \quad (10)$$

The open-loop voltage gain can now be calculated as

$$A_{VOL} = \left(\frac{V_{out}}{V_s} \right)_{OL} = A_{ROL} \times sC_s = -\frac{C_s}{C_F} \cdot \frac{G_m R_{out}}{1 + s R_{out} C'_{out}} \quad (11)$$

The closed loop voltage gain of the amplifier in Fig. 2(a) can be written as

$$A_{VCL} = \frac{V_{out}}{V_s} = \frac{A_{VOL}}{1 + LG} = \frac{-A_{VCLo}}{\left(1 + \frac{1}{\beta G_m R_{out}}\right) + s \frac{C'_{out}}{\beta G_m}} \quad (12)$$

where $A_{VCLo} = C_s/C_F$ is the ideal closed-loop DC gain of the inverting amplifier. From (12), the closed-loop bandwidth is given by

$$BW_{CL} = \frac{1}{2\pi} \frac{\beta G_m}{C'_{out}} \quad \beta \times w_{uLG} = BW_{CL} \quad (13)$$

and the closed loop DC gain is

$$A_{VCLdc} \approx -A_{VCLo} \times \left(1 - \frac{1}{\beta G_m R_{out}}\right) = -A_{VCLo} \times (1 - \epsilon_s) \quad (14)$$

where ϵ_s is the static gain error

$$|\epsilon_s| \approx \frac{1}{\beta G_m R_{out}} = \frac{1}{LG @ DC} \quad (15)$$

The total RMS integrated output noise in the amplification phase can be written as

$$V_{n,amp,o,rms}^2 = \frac{4kT\gamma_{eff}}{G_m} \cdot A_{VCLdc} \cdot \frac{2\pi BW_{CL}}{4} \approx \frac{kT\gamma_{eff}}{\beta C'_{out}} \quad (16)$$

where k is Boltzmann constant, T is the temperature in kelvin, and γ_{eff} is the effective noise coefficient which is given by

$$\gamma_{eff} = 2 * (\gamma_{1,2} + \gamma_{7,8} \frac{g_{m7,8}}{g_{m1,2}}) \quad (17)$$

where the factor of two captures noise from both differential branches. The input-referred noise contribution from the sampling phase is given by [8]

$$V_{n,smp,i,rms}^2 = \frac{KT}{C_s} \left(1 + \frac{C_F + C_{in}}{C_s}\right) \quad (18)$$

Thus, the total input referred RMS integrated noise is

$$V_{n,i,rms}^2 = \left(\frac{KT C'_{in}}{C_s^2}\right) \left(1 + \frac{\gamma_{eff}}{1 - \beta + \frac{C_L}{C_F}}\right) \quad (19)$$

IV. DESIGN PROCEDURE

The design procedure starts by determining the drain-source voltage (V_{DS}) and the source-bulk voltage (V_{SB}) of each transistor in order to obtain accurate values from the lookup tables. It is assumed that in all PMOS transistors source is connected to bulk, which means transistors M1 to M4 have $V_{SB} = 0$. In addition, for M7,8 the source is connected to ground. This leaves us with M5,6 with source-bulk voltage given by

$$V_{SB5,6} = V_{DS7,8} \quad (20)$$

Using output swing specification (V_{swing}), it is fair to split the headroom equally between transistors in the stack. With five transistors in the stack, we determine the minimum drain-source voltage of each transistor ($V_{DS,min}$) when output swing is maximum as follows

$$V_{DS,min} = \frac{V_{DD} - \frac{V_{swing}}{2}}{5} = 0.24V \quad (21)$$

To achieve the output swing specification, the output common-mode voltage ($V_{out,CM}$) should be placed at the middle of the output swing range

$$V_{out,CM} = 2V_{DS,min} + \frac{V_{swing}}{4} = 0.78V \quad (22)$$

Cascode transistors (M3,4 and M5,6) consequently have

$$V_{DS3,4} = V_{DS5,6} = V_{DS,min} + \frac{V_{swing}}{4} = 0.54V \quad (23)$$

The drain current (I_D) of all transistors is simply half the OTA biasing current (I_{BIAS})

$$I_D = \frac{I_{BIAS}}{2} = 10\mu A \quad (24)$$

(A) Design of the Input Pair (M1,2)

The design of the input-pair represents the biggest challenge as it affects most of the amplifier specifications. First, we start with the noise specification. The overall noise performance of cascaded SC circuits is optimum if the noise contributions from the sampling phase and the amplification phase are made equal [2] [9]. Thus, (19) is rearranged to obtain (C_S) as follows,

$$\gamma_{eff} = 1 - \beta + \frac{C_L}{C_F} \quad (25)$$

$$C_S = \frac{2KT \left(1 + \frac{1}{A_{VCLo}} + \frac{C_{in}}{C_S} \right)}{V_{n,i,rms}^2} \quad (26)$$

Then, modify (13) to get $(g_m/I_D)_{1,2}$ as follows

$$\left(\frac{g_m}{I_D} \right)_{1,2} = \frac{2\pi C_{out} BW_{CL}}{\beta I_D} \quad (27)$$

Next, to simplify the procedure, we assume the intrinsic gain of the cascode transistors (M3,4 and M5,6) and the input pair (M1,2) are equal. We also assume the output resistance of the current source load (M7,8) equals that of the input pair (M1,2). The previous assumptions can be written as

$$\left(\frac{g_m}{g_{ds}} \right)_{1,2} = \left(\frac{g_m}{g_{ds}} \right)_{3,4} = \left(\frac{g_m}{g_{ds}} \right)_{5,6} \quad (28)$$

$$g_{ds1,2} = g_{ds7,8} \quad (29)$$

From (3) and (15), we get the intrinsic gain of the input pair

$$\left(\frac{g_m}{g_{ds}} \right)_{1,2} = \sqrt{\frac{2}{\beta \epsilon_s}} \quad (30)$$

The input pair can be designed from (26), (27) and (30), with only one obstacle that C_{in} exists in each equation. Here comes the power of automation, which enables an iterative procedure even if a closed form solution does not exist. Simply put, we begin with $C_{in} = 0$ to have an initial solution, which gives a new C_{in} value. This loop is repeated till convergence is achieved as shown in Fig. 4 (a).

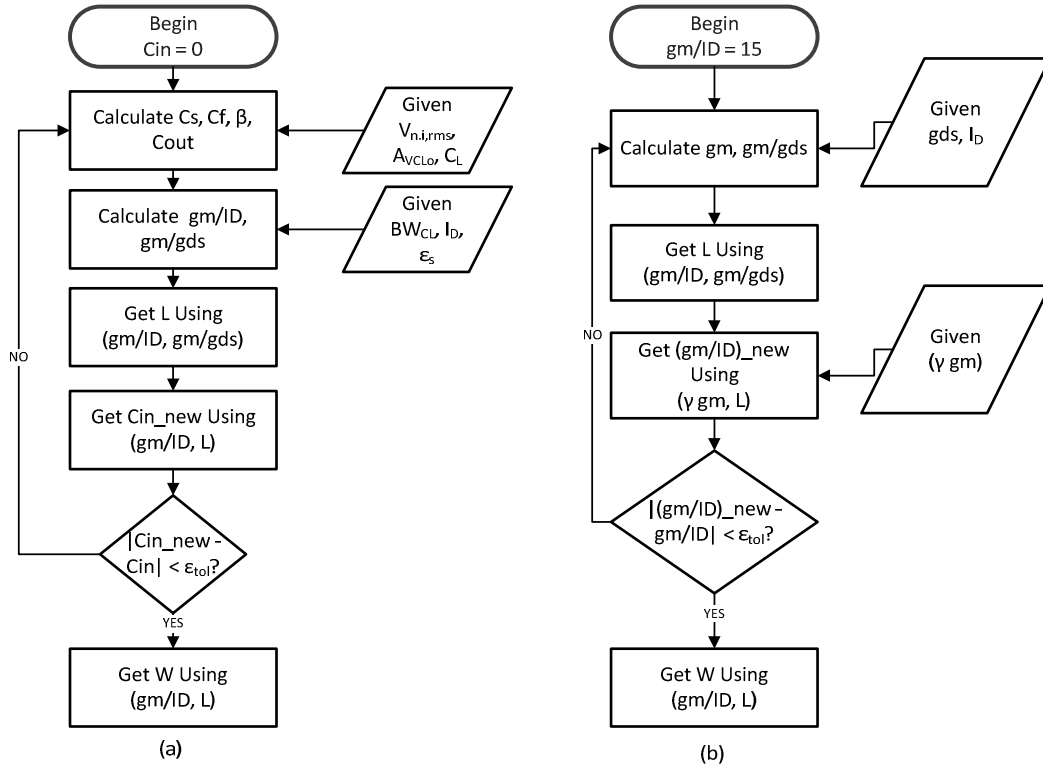


Fig. 4: Design flow of (a) the input pair (M1,2) and (b) the load (M7,8)

In detail, we use $C_{in} = 0$ to calculate the initial values of C_s, C_f, β , and C'_{out} . Then, compute $(g_m/I_D)_{1,2}$ and $(g_m/g_{ds})_{1,2}$ from (27) and (30), respectively. Next, use the computed values to get the length of M1,2 ($L_{1,2}$) from the lookup tables. Now, we have initial values of $(L_{1,2})$ and $(g_m/I_D)_{1,2}$ to use them to re-solve (4) and get improved guess for C_{in} . The updated value of C_{in} will not be equal to zero; thus, we go back to (26), get new (C_s), and repeat the previous steps in a loop. When the iterations converge with acceptable tolerance (i.e. $\epsilon_{tol} = 1\%$) in C_{in} , the final $L_{1,2}$ and $(g_m/I_D)_{1,2}$ define the current density $(I_D/W)_{1,2}$ and subsequently the width ($W_{1,2}$). The loop results in $C_s = 1.77pF$ and $C_f = 0.885pF$.

(B) Design of the Current Source Load (M7,8)

Equation (29) constrains the current source load output conductance ($g_{ds7,8}$) to be equal to that of the input pair, which is already designed in the previous subsection

$$g_{ds7,8} = g_{ds1,2} = 2.36\mu S \quad (31)$$

Next, suppose that g_{ds} is given by

$$g_{ds} = \lambda I_D \quad (32)$$

where λ is the channel length modulation coefficient [10], and for now it will be assumed that it is not a function of (g_m/I_D) (this assumption will be revisited shortly). Thus, we can choose any value for $(g_m/I_D)_{7,8}$ to estimate the length ($L_{7,8}$) from (g_m/g_{ds}) tables as follows

$$\left(\frac{g_m}{I_D}\right)_{7,8} = 15 \text{ V}^{-1}, \left(\frac{g_m}{g_{ds}}\right)_{7,8} = \left(\frac{g_m}{I_D}\right)_{7,8} * \frac{I_D}{g_{ds7,8}} = 63.56 \quad (33)$$

$$L_{7,8} = 0.73\mu m \quad (34)$$

A second constraint that defines our design procedure is in (25) and (17). Since the input pair is designed, subsequently, noise coefficient ($\gamma_{1,2}$) and transconductance ($g_{m1,2}$) are determined from the lookup tables. Certain (γg_m) is required for load which is met at a specific (g_m/I_D)

$$\gamma_{eff} = 1 - \beta + \frac{C_L}{C_F} = 2.94 \quad (35)$$

$$\gamma_{eff} = 2 * \left(\gamma_{1,2} + \gamma_{7,8} * \left(\frac{g_{m7,8}}{g_{m1,2}} \right) \right) \quad (36)$$

$$(\gamma g_m)_{7,8} = 88.02 \mu S \quad (37)$$

$$\left(\frac{g_m}{I_D} \right)_{7,8} = 10.12 V^{-1} \quad (38)$$

Back to (32), a claim that relies on the square law model, which is not completely accurate. However, the charm of automation comes with a simple iteration. As shown in Fig. 4 (b), and iterative procedure is used to retrieve accurate values of ($L_{7,8}$) and (g_m/I_D)_{7,8}. The iterations converge quickly as γ is a weak function of L . The initial and final values for the iterative procedure are shown in Table 2.

Table 2: Current source load (M7,8) parameters

Initial Values	$gm/ID (V^{-1})$	gm/gds	$L (\mu m)$
	15	63.56	0.73
Final Values	$gm/ID (V^{-1})$	gm/gds	$L (\mu m)$
	10.15	43.15	1.49

(C) Design of the Cascode Transistors (M3,4 and M5,6)

The intrinsic gain (g_m/g_{ds}) is set according to (28) to be the same as the input pair; thus, only g_m/I_D is required to complete the design. Output swing forces a min V_{DS} on the cascode pairs calculated in (21). As long as the drain-source saturation voltage (V_{DSAT}) is below this value, both input cascode and load cascode are guaranteed to operate in saturation. Provided that V_{DSAT} hardly depends on length, assume a suitable value for length (e.g., $L = 1 \mu m$) to estimate a bottom limit for g_m/I_D . However, the intrinsic gain deteriorates when V_{DS} is close to V_{DSAT} ; thus, the transistor should not be operated at the edge of saturation. For the given specifications, $g_m/I_D = 13 V^{-1}$ is chosen, which leads to a lower V_{DSAT} than the maximum allowed. Using this later (g_m/I_D) value, the rest of the parameters are found similar to the previous subsections.

V. RESULTS AND VERIFICATION

The previous design procedure is written as a MATLAB program to automate the design process. For the set of specifications given in Table 1, the sizing results are shown in Table 4. Moreover, the MATLAB program generates Spectre netlists for the OTA and the verification test bench. The developed MATLAB program uses MATLAB RF/Spectre Toolbox to call Spectre and run various types of simulations. The simulation results are retrieved and analyzed inside MATLAB. A summary of Spectre simulation results is shown in Table 5, where it is clear that the results are very close to the required specifications.

Table 4: Final Design Parameters

Transistor	Type	$L(\mu m)$	$W(\mu m)$	$g_m(\mu S)$	$r_o(k\Omega)$	g_m/g_{ds}	g_m/I_D
Input pair (M1,2)	PMOS	0.61	13.86	151	424	63.9	15.1
Input cascode (M3,4)	PMOS	0.37	5.26	130	479	93.9	13.0
Load cascode (M5,6)	NMOS	0.31	1.11	130	481	64.1	13.0
Current source load (M7,8)	NMOS	1.59	3.73	101	435	44.2	10.2

Table 5: Cadence Spectre simulation results.

OTA Current Consumption	20.0 μ A
RMS Noise Voltage	85.9 μ Vrms
Closed-Loop DC Gain	1.99
Closed-Loop Bandwidth	2.99 MHz
Static Gain Error	0.15%
Open-Loop DC Gain	66.7 dB

VI. CONCLUSION

A MATLAB automation program for the design of a switched capacitor amplifier was proposed. The amplifier uses a fully differential Telescopic cascode OTA. Multiple iterative loops are used to solve problems that has no closed form solutions. The automation results are verified using Cadence Spectre simulator. Similar design procedure could be applied for different topologies, technologies, and specifications. This opens the door to new opportunities in the automation of analog circuits, which could make a leap in the productivity and complexity of the analog IC design flow. The designers will focus more on the system design rather than the circuit level, saving time and effort, and opening the door for more system level innovations.

REFERENCES

- [1] S. Seth and B. Murmann, "Settling Time and Noise Optimization of a Three-Stage Operational Transconductance Amplifier," IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 60, pp. 1168-1174, 2013.
- [2] P. G. Jespers and B. Murmann, Systematic Design of Analog CMOS Circuits, Using Pre-Computed Lookup Tables, United Kingdom: Cambridge University Press, 2017.
- [3] N. Atkinson and et. al., "The Quad-Path Hardening Technique for Switched-Capacitor Circuits," IEEE Transactions on Nuclear Science, pp. 4356 - 4361, 2013.
- [4] Z. Tan and et. al., "An energy-efficient 15-bit capacitive-sensor interface based on period modulation," Solid-State Circuits, IEEE Journal of, vol. 47, pp. 1703-1711, 2012.
- [5] H. Omran, A. Alhoshany, A. Alahmadi and K. N. Salama, "A 33fJ/Step SAR Capacitance-to-Digital Converter Using a Chain of Inverter-Based Amplifiers," IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 64, no. 2, pp. 310-321, 2017.
- [6] B. Murmann, "EE214B HSpice models and gm/ID starter kit," December 2016. [Online]. Available: <https://web.stanford.edu/~murmnn/gmid>. [Accessed Sep 2017].
- [7] C. D. Ezekwe, W. Geiger and T. Ohms, "A 3-axis open-loop gyroscope with demodulation phase error correction," in Solid- State Circuits Conference - (ISSCC), 2015 IEEE International, San Francisco, CA, USA, 2015.
- [8] B. Murmann, "Thermal Noise in Track- and- Hold Circuits: Analysis and Simulation," IEEE Solid- State Circuits Mag., vol. 4, no. 2, p. 46– 54, 2012.
- [9] D. W. Cline and P. R. Gray, "Power Optimized 13-b 5Msamples/s Pipelined Analog-to-Digital Converter in 1.2 μ m CMOS," IEEE J. Solid- State Circuits, vol. 31, no. 3, p. 294– 303, 1996.
- [10] P. R. Gray, P. J. Hurst, S. H. Lewis and R. G. Meyer, Analysis and Design of Analog Integrated Circuits, 5th ed., Wiley, 2009.